Android OS uses middleware to support the hardware. Whereas, Apple developed its iOS by shrinking the Mac OS, and as the hardware for the iPhone was designed for iOS, there were no compatibility issues of the kind Android had. The Android OS also suffers from fragmentation at some point of time. For this reason, Android smartphones have performance issues after some use. RTOSs can not only improve the performance of smartphones but also IoT devices. As the I/O response of RTOSs is better than GPOSs, IoT devices can be controlled with minimum latency.

Therefore, to get a better solution, Google is now working on the development of a new OS called Fuchsia. Google Fuchsia is an open source RTOS that is capable of running on many platforms — from embedded systems and smartphones to tablets and personal computers. As Fuchsia is an RTOS, it will not cause any latency and performance issues like the Android OS. In contrast to prior Google-developed operating systems such as Chrome OS and Android, which are based on the Linux kernel, Fuchsia is based on a new kernel called Zircon, named after the mineral. Zircon is derived from Little Kernel, a small operating system intended for embedded systems. 'Little Kernel' was developed by Travis Geiselbrecht, a creator of the NewOS kernel used by Haiku. The initial development of Fuchsia has started at GitHub. On July 1, 2019, Google had announced the official website (*https://fuchsia.dev*) of the development project, providing source code and documentation for the operating system.

Requirements of Fuchsia

- Google Fuchsia can be improved and customised using various programming languages and runtimes, including C++, Rust, Flutter, and Web.
- Flutter, a software development kit, is required to code the apps and UI for Fuchsia.
- A 64-bit Intel machine with at least 8GB of RAM and 100GB of free disk space is required to develop Fuchsia.
- One of the best ways to experience Fuchsia is by running it on actual hardware like Acer Switch 12, Intel NUC, and Google Pixelbook.
- The Fuchsia install process, called 'paving', requires two connected machines over a network — the machine on which you want to run Fuchsia ('target') and the machine on which you build Fuchsia ('host').
- Fuchsia creates a UNIX-like file system called MinFS, and can currently support files up to 4GB.
- It requires AArch64 (ARM64) and x86-64 platforms to run.

Functionality and architecture

Google Fuchsia doesn't use a microkernel but uses a different type of kernel called message passing. A message passing kernel is a kind of Inter Process Communication (IPC) mechanism, where processes communicate with each other by passing messages. Unlike the shared memory kernel, which is also another kind of IPC, a message passing kernel doesn't share the memory with the processes but uses a message queue instead.

In a message passing kernel, one process can send a message to another through the message queue, and the message carries the information about the requirement. Each message that is sent by a process contains the identity of the process that is going to receive it. Similarly, the process that receives the message creates another message, sends it back to the previous process and informs about the requirement. In this way, messages are passed between every process of the kernel without any flaw.

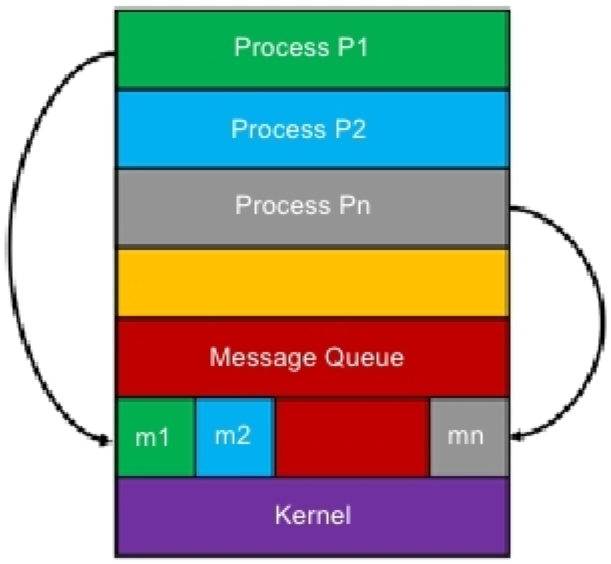

Figure 1: Message passing process

If there are two processes (P1 and P2) running in the kernel, and P1 needs to use a resource that is being currently used by the process P2, then P1 will send a message to P2 to release the resource within a specified time. As soon as P2 receives the message, it will try to complete its process within the time specified by P1, and the kernel will give higher priority to P2. After completing the process, P2 will inform P1 by passing a message that the resource is free to use, and P1 will use the resource after getting the message. This process of message passing can keep the system running without any latency, and is also easier to implement than a shared memory process.

Fuchsia has four layers, as shown in Figure 2.

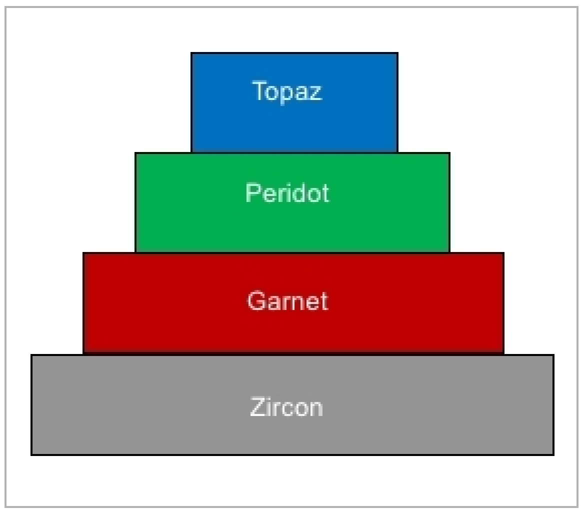

Figure 2: Fuchsia layers

Zircon: This is the first layer and is the kernel of Fuchsia. Zircon is composed of a new kind of message